Quiz No:-

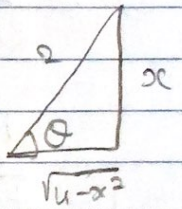
C11. $$\int_2^\infty \frac{1}{x\sqrt{4-x^2}}\,dx = \int_2^\infty \left(\frac{1}{x}\right)\left(\frac{1}{\sqrt{4-x^2}}\right)dx$$

$$= \lim_{b\to\infty} \int_2^b \left(\frac{1}{x}\right)\left(\frac{1}{\sqrt{4-x^2}}\right)dx$$

Let $x = 2\sin\theta$

$dx = 2\cos\theta\,d\theta$

$x = 2\sin\theta$



$$\int \left(\frac{1}{x}\right)\left(\frac{1}{\sqrt{4-x^2}}\right)dx$$

$$= \int \left(\frac{1}{2\sin\theta}\right)\left(\frac{1}{\sqrt{4-4\sin^2\theta}}\right)2\cos\theta\,d\theta$$

$$= \int \left(\frac{\cos\theta}{\sin\theta}\right)\frac{1}{2\cos\theta}\,d\theta$$

$$= \int \frac{1}{2\sin\theta}\,d\theta = \frac{1}{2}\int \frac{1}{\sin\theta}\,d\theta$$

$$= \frac{1}{2}\int \csc\theta\,d\theta$$

$$= \frac{1}{2}\left[\ln|\csc\theta + \cot\theta| + C\right]$$

$$= \frac{1}{2}\left[\ln\left|\frac{2}{x} + \frac{\sqrt{4-x^2}}{x}\right|\right] + C$$

$$= \frac{1}{2} \ln \left| \frac{2+\sqrt{4-x^2}}{x} \right| + C$$

$$\therefore \lim_{b\to\infty} \int_2^b \left(\frac{1}{x}\right)\left(\frac{1}{\sqrt{4-x^2}}\right) dx = \left[\frac{1}{2} \ln \left| \frac{2+\sqrt{4-x^2}}{x} \right|\right]_2^b$$

$$= \lim_{b\to\infty} \frac{1}{2} \left[\ln \left| \frac{2+\sqrt{4-b^2}}{b} \right| - \ln \left| \frac{2+\sqrt{0}}{2} \right| \right]$$

$$= \left(\frac{1}{2}\right) \left[\ln \left| \frac{\infty}{\infty} \right| - \ln |1| \right]$$

$$= \frac{1}{2}$$

$\therefore \int_2^\infty \frac{1}{x\sqrt{4-x^2}}\, dx$ is Converges.

(C12.) $y = \frac{x^3}{6} + \frac{1}{2x}$, $2 \leq x \leq 3$

$$L = \int_2^3 \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

$$y = \frac{x^3}{6} + \frac{1}{2x}$$

$$\frac{dy}{dx} = \frac{3x^2}{6} + \frac{-1}{2} x^{-2}$$

$$\frac{dy}{dx} = \left(\frac{x^2}{2} - \frac{1}{2x^2}\right)$$

$$\left(\frac{dy}{dx}\right)^2 = \left(\frac{x^2}{2} - \frac{1}{2x^2}\right)^2$$

$$\left(\frac{dy}{dx}\right)^2 = \left(\frac{x^4}{4} - \frac{2x^2}{4x^2} + \frac{1}{4x^4}\right)$$

$$L = \int_2^3 \sqrt{1 + \frac{x^4}{4} + \frac{1}{4x^4} - \frac{1}{2}}\, dx$$

$$L = \int_2^3 \sqrt{\frac{x^4}{4} + \frac{1}{4x^4} + \frac{1}{2}}\, dx$$

$$L = \int_2^3 \sqrt{\left(\frac{x^2}{2} + \frac{1}{2x^2}\right)^2}\, dx$$

$$L = \int_2^3 \left(\frac{x^2}{2} + \frac{1}{2x^2} \right) dx$$

$$L = \left[\frac{x^3}{6} + \frac{1}{2}\left(-\frac{1}{x}\right) \right]_2^3$$

$$L = \left[\frac{x^3}{6} - \frac{1}{2x} \right]_2^3$$

$$L = \left[\frac{3^3}{6} - \frac{1}{2(3)} \right] - \left[\frac{2^3}{6} - \frac{1}{2(2)} \right]$$

$$L = \left[\frac{9}{2} - \frac{1}{6} \right] - \left[\frac{4}{3} - \frac{1}{4} \right]$$

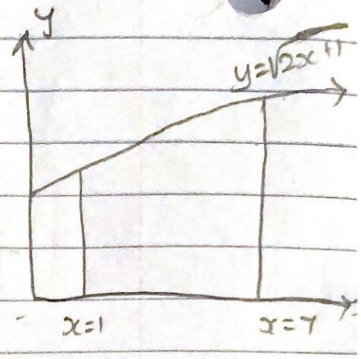
$$L = \left[\frac{9}{2} - \frac{1}{6} - \frac{4}{3} + \frac{1}{4} \right]$$

$$\boxed{L = \frac{13}{4}}$$

(13) $y = \sqrt{2x+1}$ and x-axis $1 \le x \le 7$ revolved about x-axis.

$$SA = 2\pi \int_a^b (\text{Area})(\text{Length of the solid})\, dx$$

$$SA = \int_1^7 (2\pi)(\sqrt{2x+1})\ (\text{length})\ dx$$



$$y = \sqrt{2x+1}\ , \quad y = (2x+1)^{1/2}$$

$$\frac{dy}{dx} = \frac{1}{2}(2x+1)^{-1/2} \cdot (2)$$

$$\frac{dy}{dx} = \frac{1}{(2x+1)^{1/2}} = \frac{1}{\sqrt{2x+1}}$$

$$\left(\frac{dy}{dx}\right)^2 = \left(\frac{1}{2x+1}\right)$$

$$SA = \int_1^7 (2\pi)(\sqrt{2x+1})\left(\sqrt{1+\frac{1}{2x+1}}\right)dx$$

$$= \int_1^7 (2\pi)\sqrt{2x+1}\left(\sqrt{\frac{2x+1+1}{2x+1}}\right)dx$$

$$= \int_1^7 (2\pi)\sqrt{2x+1}\left(\sqrt{\frac{2x+2}{2x+1}}\right)dx$$

$$= \int_1^7 (2\pi)(\sqrt{2x+2})\,dx$$

$$= \int_1^7 (2\pi)(\sqrt{2})(\sqrt{x+1})\, dx$$

$$= 2\sqrt{2}\pi \int_1^7 \sqrt{x+1}\, dx$$

Let $u = x+1$, $du = dx$

$$= 2\sqrt{2}\pi \int \sqrt{u}\, du$$

$$= 2\sqrt{2}\pi \left[\frac{2}{3} u^{3/2} \right]$$

$$= 2\sqrt{2}\pi \left[\frac{2}{3}(x+1)^{3/2} \right]_1^7$$

$$= 2\sqrt{2}\pi \left(\left[\frac{2}{3}(7+1)^{3/2} \right] - \left[\frac{2}{3}(1+1)^{3/2} \right] \right)$$

$$= 2\sqrt{2}\pi \left[\left(\frac{16(\sqrt{8})}{3} \right) - \left(\frac{4\sqrt{2}}{3} \right) \right]$$

$$= 2\sqrt{2}\pi \left(\frac{32\sqrt{2}}{3} - \frac{4\sqrt{2}}{3} \right]$$

$$= 2\sqrt{2}\pi \left(\frac{28\sqrt{2}}{3} \right) = \frac{112\pi}{3}$$

$$\boxed{SA = \frac{112\pi}{3}}$$